EMMANUEL M. NDALIRO

Senior Software Engineer — AI Systems, Code Evaluation & Architecture

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PROFESSIONAL SUMMARY

Senior Software Engineer with 6+ years of experience in software architecture, system design, code review, and debugging, specializing in AI/LLM systems for the past 4 years. I build the kind of AI systems this role evaluates: LLM-powered coding agents, RAG pipelines, and multi-agent orchestration with model-evaluation frameworks that score AI output quality, correctness, and reliability. Experienced at reviewing complex codebases and producing detailed written technical feedback with path-level evidence and reproducible defect cases. Published technical writer on AI agent architecture (Medium), open-source contributor (GitHub).

CORE COMPETENCIES

- **Code Review & Evaluation:** Structured review of AI-generated and human-written code for correctness, quality, and design; evaluation frameworks that score LLM/agent output against production standards
- **System Design & Software Architecture:** Microservices, agent orchestration architectures, human-in-the-loop approval systems, API design, event-driven pipelines
- **Debugging:** Root-cause analysis in complex codebases; reproduced subtle production defects offline with path:line evidence and automated test harnesses
- **Algorithms & Data Structures:** Applied in production ranking/scoring engines (weighted scoring, taxonomy resolution, deduplication, state machines) and LLM pipelines (retrieval, chunking, embedding search)
- **AI/ML Systems:** LLM integration and orchestration, RAG / Agentic RAG, model fine-tuning (Hugging Face, Ollama), prompt engineering, model evaluation and validation pipelines
- **Technical Writing:** Published author of in-depth technical analyses; expert-level written English for feedback, rubrics, and documentation

PROFESSIONAL EXPERIENCE

Software Developer — AHADI Corporation · Nairobi, Kenya

Oct 2023 – Present

- Develop model evaluation frameworks that assess AI output quality, correctness, and reliability — scoring LLM-generated code and text against production-grade standards with structured test cases
- Architect microservices with Flask and Django in containerized environments with CI/CD pipelines across AWS, Azure, and GCP
- Build LLM-powered applications using RAG, agentic workflows, and prompt engineering, improving automated response quality for production services
- Implement comprehensive monitoring and logging, enabling proactive issue detection and reducing incident response time
- Review and debug complex service codebases; produce detailed written technical assessments with evidence-backed recommendations

AI/ML Engineer — Independent / Freelance · Remote

Sep 2023 – Present

- Engineer AI agents with Python, Pydantic-AI, and LangGraph using structured outputs; published implementations and technical analyses on GitHub and Medium, including "The AI Protocol Wars: MCP vs. A2A" and "Building Structured AI Agents"
- Train and fine-tune LLMs using Hugging Face Transformers and Ollama; build validation pipelines with open-source datasets to evaluate and improve model performance

- Integrate LLM, RAG, and agent systems with cloud-native backends (Flask, Django, FastAPI), reducing manual processing time by 40% for client workflows
- Design and ship a production AI job-search platform end-to-end: Flask REST API (59 endpoints), Next.js/TypeScript dashboard, Playwright browser automation, and GitHub Actions CI/CD — including the code-review, debugging, and architecture decisions across the full stack
- Audit third-party codebases for correctness, security, and architectural quality; document defects with path:line evidence and offline reproductions

Python Backend Developer — Freelance / Contract · Remote

Jul 2021 – Dec 2022

- Developed Python automation scripts and ML pipelines; built cross-platform Flutter applications consuming AI APIs
- Managed Flask/Django backends with Firebase/SQLite databases and RESTful API integrations
- Automated end-to-end test and scraping workflows with Selenium and Playwright

Mobile Developer & Backend Engineer — Contract · Remote

May 2020 – Jun 2021

- Developed Python backends for mobile applications, integrating cloud services and Google Maps APIs
- Built cross-platform mobile apps with Flutter/Dart, deployed to Firebase and cloud-hosted services
- Collaborated with UX designers to convert wireframes into functional, polished product experiences

TECHNICAL SKILLS

Software Engineering: System Design · Software Architecture · Code Review · Debugging · Algorithms · Data Structures · Microservices · RESTful APIs · Design Patterns · Testing & Test-Case Design

AI/ML: LLM Evaluation & Orchestration · RAG / Agentic RAG / Hybrid RAG · AI Agents (LangChain, LangGraph, Pydantic-AI) · Fine-tuning (Hugging Face, PyTorch, TensorFlow) · Prompt Engineering · Model Evaluation · MCP / A2A Protocols · Ollama

Backend: Python (Flask, Django, FastAPI) · PostgreSQL · Firebase · SQLite · Docker · CI/CD (GitHub Actions)

Cloud & DevOps: AWS · Azure · GCP · Serverless (Lambda, Cloud Functions) · Infrastructure as Code · Monitoring & Logging

Frontend & Automation: React · Next.js · TypeScript · Flutter (Dart) · WebSockets · Playwright · Selenium · Web Scraping

EDUCATION

BSc. Software Engineering — Multimedia University of Kenya · 2015–2019 · Second Class Honours

Relevant coursework: Data Structures & Algorithms · Software Architecture & Design · Database Systems · Software Testing & Quality Assurance

CERTIFICATIONS

Google Africa Developer Training Program — Advanced Cloud & Development Track

Test Automation University:

- Foundation for Test Automation
- Python Programming
- Source Control for Test Automation
- Automated Testing
- Data Science

PUBLICATIONS & OPEN SOURCE

Technical Publications (Medium — medium.com/@kram254)

- *"The AI Protocol Wars: MCP vs. A2A — A Comprehensive Analysis with Practical Implementations"* — in-depth comparative analysis of AI agent communication protocols, with working code evaluating interoperability, reliability, and design trade-offs
- *"Building Structured AI Agents: Harnessing DeepSeek R1 and Pydantic AI"* — hands-on engineering guide to building agents with structured, validated outputs

Open Source (GitHub — github.com/kram254)

- Author of open-source Python agent frameworks and automation tools — including AI-powered web-scraping systems and multi-agent orchestration implementations accompanying the published articles